# PROJECT-BASED LEARNING

## PHASE-II EVALUATION

### Where's My Bus: A Hardware Independent Bus Tracker

Team Null Pointer — Team ID: DSCPP-III-2026-T007

**Department of Computer Science & Engineering**
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

# Team & Mentor Details

## Team Information

**Team ID:** DSCPP-III-2026-T007
**Team Name:** Null Pointer
**Semester:** 3rd
**Sections:** ML1, F, ML2, B
**Domain:** Tech.

## Team Members

1. Devansh Dangwal (Team Lead) – 2510370109
2. Aryan Gusain – 2510012634
3. Vinayak Lakhera – 2510010715
4. Gautam Singh – 2510012593

## Mentor

Mr.Kanishka Bhatt

**Interactions completed: 4**

# Problem Statement & Finalized Objectives

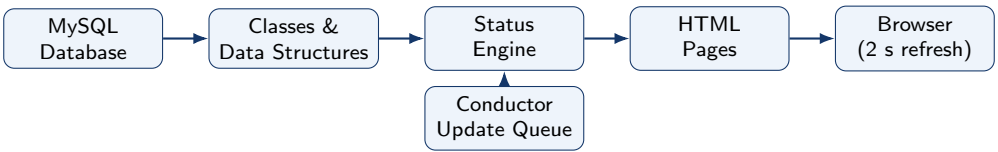
## Problem Statement

Colleges with fixed bus routes give students no easy way to know whether their bus is late, full, or out of service. *Where's My Bus* gives every student a live colour status for each bus at their own stop, without GPS hardware or paid APIs.

## Finalized Objectives

1. Show a colour-coded status for every bus at the student's chosen stop, with no login required.
2. Let the conductor update the bus position with one action at each stop.
3. Give the Head of Department (HOD) a summary report of all buses and recent updates.
4. Stay hardware independent and low cost: all routes, stops, buses and ETAs come from MySQL, built on hand-written data structures.

# Updated System Architecture / Workflow




## Major Components

- **Database:** stops, routes, order, buses, ETAs (MySQL)
- **Classes & data structures:** route = linked list; hash tables; queue and stack for updates; heap for sorting
- **Status engine:** moves bus along route, sets colour
- **Display:** C++ writes the HTML pages

## Phase-II Design Decisions

- Each layer talks only to the next; nothing is hard-coded, all data loads from MySQL at start-up
- C++ runs the MySQL command-line client in the background and reads its output
- Colours are computed live and never stored
- Pages are written to a temp file, then moved, so the browser never sees a half-written page

# Module Design & Functional Requirements

## Major Modules

1. Database & loader (25 stops, 3 routes, 14 buses)
2. Core classes & data structures (Stop, Route, Bus, Update, users)
3. Status engine (movement, colours, undo/redo, trip completion)
4. Interfaces (terminal menu, student and HOD pages)

## Functional Requirements

- Student views bus colour at a chosen stop without logging in
- Conductor logs in, starts a shift, marks next stop reached, marks seats full or maintenance, and can undo
- HOD gets a report of buses and recent updates
- Pages refresh every two seconds so changes appear immediately

## Module Dependency

Interfaces → status engine → core classes → database (loaded at start-up). Conductor updates enter a queue, are validated, then go onto the history stack used for undo.

# Technology Stack & Development Methodology

## Technology Stack

- Programming: C++ (with STL)
- Database: MySQL 8
- Framework: none (plain HTML/CSS pages)
- Development Tool: g++ (Windows build file)
- Version Control: Git / GitHub

## Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

# Implementation of PBL Course Concepts

| Course | Concepts Actually Implemented | Project Module |
|---|---|---|
| **Data Structures in C** | Singly linked list (route as ordered stops), queue (conductor updates), stack (update history and undo/redo), binary heap (bus sorting), hash table (bus and route lookup) | Core classes, status engine |
| **OOPs with C++** | Classes, inheritance (User → Student / Conductor / HOD), class templates (Queue, Stack, Heap), STL, single-owner object management | User classes, registries, data structures |
| **Operating Systems** | Process creation: MySQL command-line client run from C++ as a background process, with its output read back | Database connection layer |
| **DBMS** | SQL schema for stops, routes, stop order, buses and ETAs; custom SQL query to get correct ETAs | Database, data loader |

# Development & Implementation Progress

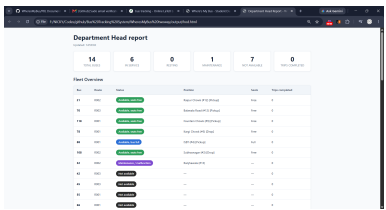
| Module | Status | Work Completed / Planned |
|---|---|---|
| Database | Completed | 25 stops, 3 routes, 14 buses and ETAs between stops (can be extended) |
| C++–MySQL link & import | Completed | Connection through the MySQL client; all data loaded at start |
| Core classes & data structures | Completed | Stop, Route, Bus, Update, user classes, queue, stack, heap, hash tables |
| Status engine (drop trip) | Completed | Bus movement, colours, trip completion, undo and redo |
| Student & HOD pages | Completed | HTML/CSS pages with 2-second auto refresh (drop trip only) |
| Conductor terminal menu | Completed | Start shift, next stop, seats, maintenance, undo |
| Pickup, ETA, dashboard | Pending | Doubly linked list for return leg; ETA; web dashboard |

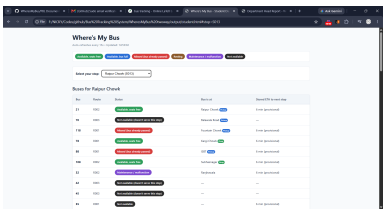
**Overall Project Completion:** 70%

On schedule: drop-trip demo. Behind: pickup tracking, ETA, dashboard, manual, report.

# Working Prototype / Demonstration

**Implemented Features:** Live colour status of drop trips • Conductor menu (start, next stop, seats, undo) • HOD reports • Auto-refreshing pages



# Testing, Results & Validation

| Test / Module | Expected Result | Observed Result | Status |
|---|---|---|---|
| Compilation check | No warnings | No warnings | Pass |
| Memory check | No leaks | No memory leak | Pass |
| Loading from MySQL | All routes and buses loaded | 3 routes, 14 buses loaded | Pass |
| Conductor operations (drop trip) | All shift operations work | All performed successfully | Pass |
| Student page colours | Colours match bus state | Colours matched expected values | Pass |
| HOD report | Report shows completed trip | Shows the successful trip | Pass |
| Data structures | Heap order, list integrity, undo correct | All checks passed | Pass |
| Error messages | Clear error messages | Wrong password, missing config, bad database shown | Pass |
| Pickup tracking | Bus returns from rest stop to Graphic Era | Bus cannot return (forward-only list) | Fail |

## Results / Validation

8 of 9 test groups pass. The one failure, pickup tracking, comes from the forward-only route linked list and will be fixed with a doubly linked list.

# GitHub Repository & Development Activity

## Repository Information

- GitHub Link: `https://github.com/Nickyyayy/WheresMyBus`
- README.md: updated
- Branches: `main`, plus one branch per member
- Folders: `backend` (C++), `frontend` (CSS), `sql`, `output` (generated pages)
- Database password kept in a separate config file, not uploaded

## Important Commits

- `e9452e9` (HEAD, main): registries
- `4250780`: data structures
- `c012719`: folder structure, Windows build file, SQL load and config, core classes
- `a3cee68`: MySQL database and README update

**Individual commit counts:** *no. of total commits/4*

# Individual Contribution

| Member | Role | Technical Contribution | GitHub Commits |
|---|---|---|---|
| Devansh Dangwal | Team Lead, core logic | Stop, Route (linked list), Bus and Update classes; status engine with bus movement, colours, trip completion, undo and redo; slides and report | *25%* |
| Aryan Gusain | Interface & integration | Conductor/HOD terminal menu, configuration and main program; HTML/CSS pages; frontend–backend connection; slides and report | *25%* |
| Vinayak Lakhera | Database & data layer | MySQL database (25 stops, 3 routes, 14 buses, ETAs); C++–MySQL connection and import; User, Student, Conductor, HOD classes; hash tables; route validation | *25%* |
| Gautam Singh | Data structures & pages | Queue, Stack and Binary heap template classes and bus sorting; student and HOD HTML/CSS pages | *25%* |

# Challenges, Remaining Work & Phase-III Roadmap

## Challenges & Solutions

- Graphic Era is first and last stop → track by stop order number, not name
- ETA view wrong for last record → own query
- C++–MySQL libraries failed on Windows → run the MySQL client from C++; quote paths with spaces
- Browser loaded half-written pages → temp file, then move
- Pointer/memory bugs → one owner per object, memory-checker tests
- One shared undo stack → pop other buses' newer updates, undo ours, push them back in order
- Pickup fails (forward-only route list) → doubly linked list planned

## Phase-III Roadmap

1. Pickup tracking with a doubly linked list (Devansh, Vinayak)
2. Conductor web dashboard (Gautam, Aryan); ETA calculation (Devansh)
3. Larger database: more routes, stops and buses (Vinayak)
4. More tests: missed bus, resting bus, wrong updates; test on the demo laptop; code review (all)
5. User manual with screenshots, final report and final presentation

# Expected Outcomes & References

## Expected Final Outcomes

- Working tracker for both drop and pickup trips, with conductor web dashboard and ETAs
- Source code, SQL files, user manual, final report and demo
- Cost-effective college bus tracking with no GPS hardware or paid API
- Easily extended: more routes, stops and buses can be added through the database

## References


1. MySQL 8.0 Reference Manual, dev.mysql.com/doc
2. C++ Standard Library reference, cppreference.com
3. Project repository: github.com/Nickyyayy/WheresMyBus


**Thank You**
Questions & Suggestions